Package ‘iRamat’

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Type Package

Title Statistical analysis on Archaeomaterial

Description Statistical analysis on Archaeomaterial.

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URL https://github.com/iramat/iRamat

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NeedsCompilation no

Author Thomas Huet [aut, cre] (<https://orcid.org/0000-0002-1112-6122>)

Maintainer Thomas Huet <thomas.huet@cnrs.fr>

Contents

chrono *Create a timeline of a selected dataset*

Description

This function parses EDTF-formatted dates from a dataset and returns one or two ggplot2 timelines: a site-based timeline and, optionally, a PeriodO timeline for comparison.

Usage

```
chrono(
  d,
  use_periodo = FALSE,
  periodo_authority = "https://n2t.net/ark:/99152/p02chr4",
  seriated = TRUE,
  all_dates = FALSE,
  time_match = 0.9,
  verbose = TRUE
)
```

Arguments

d	A data frame containing at least a column ‘edtf‘ with EDTF dates and a column ‘context_name‘.
use_periodo	Logical, if TRUE (default FALSE), also fetch and plot periods from a PeriodO authority.
periodo_authority	A string. URL of a PeriodO authority (default: "https://n2t.net/ark:/99152/p02chr4").
seriated	Logical, if TRUE (default), reorder sites/periods by earliest start date.
all_dates	Logical, if TRUE, include sites with missing/NA dates (default: FALSE).
time_match	Numeric between 0 and 1. Minimum proportion of temporal overlap required between dataset range and a period to include the period (default: 0.9).
verbose	Logical, if TRUE (default), print progress messages.

Value

A list of ggplot2 objects with elements:

sites A timeline of dataset sites (always returned).

periodo A PeriodO timeline, if ‘use_periodo = TRUE‘, otherwise NULL.

Examples

```
## Not run:
df <- db_api_connect()
plots <- chrono(df$dataset_adisser17, use_periodo = TRUE)
ggpubr::ggarrange(plots$sites, plots$periodo, heights = c(1,2), ncol = 1, align = "v")

## End(Not run)
```

`db_api_connect` *Connect the CHIPS DB API and return an R object (dataframe, etc)*

Description

Connect the CHIPS DB API and return an R object (dataframe, etc.). The default dataset is dataset_adisser17, accessed at <http://157.136.252.188:3000/dataset_adisser17>.

If `all_datasets` = `TRUE`, all datasets whose `description` is "Dataset" in the GitHub-hosted table https://github.com/iramat/iramat-dev/blob/main/dbs/chips/urls_data.tsv are downloaded (using their `url_data` column), merged into a single `data.frame`, and a new column `dataset_name` is added to indicate the source dataset for each row.

Usage

```
db_api_connect(
  d = NA,
  api_url = "http://157.136.252.188:3000/dataset_adisser17",
  api_list =
   "https://raw.githubusercontent.com/iramat/chips/hugo-files/static/data/urls_data.tsv",
  output_format = "dataframe",
  verbose = TRUE,
  all_datasets = FALSE
)
```

Arguments

`d`	A hash object. If none is provided, a new one will be created. Ignored when `all_datasets` = `TRUE`.
`api_url`	An URL landing to an API. Default: `"http://157.136.252.188:3000/dataset_adisser17"`. Ignored when `all_datasets` = `TRUE`.
`api_list`	An URL landing to a TSV file listing API URLs. Default: `"https://raw.githubusercontent.com/` Ignored when `all_datasets` = `FALSE`.
`output_format`	The selected output format. Default "dataframe" (currently not used).
`verbose`	if TRUE (by default), verbose.
`all_datasets`	Logical. If `TRUE`, download all datasets with `description == "Dataset"` listed in the GitHub table https://github.com/iramat/iramat-dev/blob/main/dbs/chips/urls_data.tsv, merge them into a single `data.frame`, and add a column `dataset_name` with the name of the source dataset (last component of the API URL). Default `FALSE`.

Value

* If `all_datasets` = `FALSE` (default): a hash where each key is the dataset label (last part of `api_url`) and each value is a dataframe. * If `all_datasets` = `TRUE`: a single merged `data.frame` with an additional column `dataset_name`.

Examples

```
# Default behaviour: one dataset stored in a hash
df_hash <- db_api_connect()
head(df_hash$dataset_adisser17)

# All CHIPS datasets listed in urls_data.tsv, merged into a single dataframe
df_all <- db_api_connect(all_datasets = TRUE)
head(df_all)
```

---

periodo — *Create a timeline of periods from a PeriodO authority*

Description

This function reads a PeriodO authority file (CSV + JSON metadata) and returns a ggplot2 timeline of the defined periods. Optionally, the timeline can be restricted to a given temporal range, and periods can be filtered by the amount of overlap with that range.

Usage

```
periodo(
  periodo_authority = "https://n2t.net/ark:/99152/p02chr4",
  use_periodo = FALSE,
  min_date = NA,
  max_date = NA,
  time_match = NA,
  location = NA,
  seriated = TRUE,
  verbose = TRUE
)
```

Arguments

`periodo_authority`	A string. URL of a PeriodO authority (default: "https://n2t.net/ark:/99152/p02chr4", INRAP).
`use_periodo`	Logical, if TRUE, the function is used as a subroutine (typically called from [chrono()]) and the plot title is suppressed.
`min_date`	Optional numeric. Lower bound of the time interval (most ancient date). Default: ‘NA‘ (no lower bound).
`max_date`	Optional numeric. Upper bound of the time interval (most recent date). Default: ‘NA‘ (no upper bound).
`time_match`	Numeric between 0 and 1. Minimum proportion of temporal overlap required between a period and the given [min_date, max_date] interval for the period to be included. Default: ‘NA‘ (no filtering).
`location`	Optional string. If provided, only periods whose ‘spatial_coverage‘ field contains this string will be included (e.g. "France"). Default: ‘NA‘ (no filtering).
`seriated`	Logical, if TRUE (default), reorder periods by earliest start date.
`verbose`	Logical, if TRUE (default), print progress messages.

Value

A list with one ggplot2 object:

periodo A timeline of PeriodO periods.

Examples

```
## Not run:
# Default authority (INRAP)
periodo()

# Restrict to a specific interval and require exact overlap
periodo(min_date = -700, max_date = 0, use_periodo = TRUE, time_match = 1)

# Authority ArkeOpen, France only
periodo(periodo_authority = "http://n2t.net/ark:/99152/p09hq4n", min_date = -500, max_date = 500, use_periodo

## End(Not run)
```

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ppa *Classify a Point Pattern Distribution*

Description

Classifies a point pattern distribution from raster images using common geostatistical tests.

Usage

```
ppa(
  d = NA,
  root = "https://raw.githubusercontent.com/iramat/iRamat/master/inst/extdata/",
  img.paths = c("clustered_distribution.png", "random_distribution.png",
    "regular_distribution.png"),
  ppa_tests = c("quadrat", "ripley", "gfunction"),
  verbose = TRUE
)
```

Arguments

d	A hash object. If none is provided, a new one will be created.
root	A character string specifying the path to the raster dataset.
img.paths	A character vector containing the names of the images to be analyzed.
ppa_tests	A character vector specifying the point pattern analysis tests to perform. Supported options are: Quadrat Test (‘"quadrat"‘), Ripley’s K-function (‘"ripley"‘), and G-function (‘"gfunction"‘). By default: ‘c("quadrat", "ripley", "gfunction")‘. **Interpretation guidance**: - ‘quadrat.test()‘: if *p* < 0.05 → the pattern is likely non-random (clustered or regular) - ‘Kest()‘ (Ripley’s K): analyzes spatial clustering at multiple scales - ‘Gest()‘ (G-function): focuses on nearest-neighbor distances
verbose	Logical. If ‘TRUE‘ (default), prints informative messages during processing.

Value

A list or object containing the results of the specified point pattern analysis tests.

Examples

```
d <- ppa()
# Show results of Quadrat test of the clustered distribution
d$clustered_distribution.png$quadrat
# Plot the K-Ripley of the clustered distribution
plot(d$clustered_distribution.png$ripley, main = "clustered distribution")
# Plot the G-function of the regular distribution
plot(d$regular_distribution.png$gfunction, main = "regular distribution")
```

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